

ALI TURAB

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SUMMARY

Full Stack AI Engineer with expertise in Machine Learning, MERN Stack, and TypeScript, specializing in developing AI-integrated full-stack applications with strong focus on performance and scalability. Passionate about cutting-edge technologies, building RAG systems and agentic systems, leveraging Deep Learning and Large Language models (LLMs), to enhance software intelligence, automation, and user-centric decision-making.

EXPERIENCE

Synviz Solutions - Associate Software Engineer (ASE)
 September 2025 – March 2026

- Designed and implemented a production-grade RAG system using Qwen2.5, improving retrieval relevance by ~35% via enhanced embedding alignment and query-context optimization.
- Developed and fine-tuned transformer-based systems using DeBERTa-v3 and RoBERTa-large for text classification and natural language understanding, improving accuracy across domain-specific datasets.
- Built and trained CNN-based vision pipelines using ConvNeXt and DenseNet for image classification and feature extraction.
- Designed and built scalable backend systems using FastAPI and Node.js, focusing on high-performance microservices and reliability.

NETSOL Technologies - AI/ML Intern
 July 2025 – August 2025

- Built RAG-based financial intelligence system (LangChain, OpenAI API) over 5K+ synthetic product dataset, enabling personalized savings recommendations and contextual financial insights.
- Developed time-series forecasting + investment models using LSTM on 10-year Vanguard S&P 500 ETF data, combined with regression pipelines and real-time market/news ingestion for signal generation.
- Engineered full-stack financial platform integrating this architecture with FastAPI services and React + TypeScript frontend for scalable decision workflows.
- Implemented OAuth 2.0 and optimized frontend rendering efficiency, improving responsiveness across core user flows.

AIIESEC in Pakistan:

- Out-Going Volunteer Team Member
 March 2024 - August 2024
- Business Development Team Member
 September 2023 - March 2024

EDUCATION

COMSATS University, Lahore
 September 2022 – June 2026

Bachelor in Software Engineering
 CGPA: 3.69/4

Punjab Group of Colleges, Main Campus, Lahore
 2020-2021

HSSC Marks: 1020/1100 (92%)

Garrison Academy DHA Campus, Lahore
 2019–2020

SSC Marks: 1059/1100 (96%) | Gold Medalist

SKILLS

- Programming Languages:** Java, C++, C, JavaScript, Type Script, Python
- Frontend Development:** HTML, CSS, React.js, Next Js, Tailwind CSS, Bootstrap, ShadCN, Material UI
- State Management:** Redux Toolkit, Zustand
- Backend Development:** Node.js, Express.js, FastAPI
- Databases:** PostgreSQL, MongoDB, MS SQL Server, Convex DB, Vector Databases(Chroma DB, Pinecone)
- Machine Learning & Deep Learning:** PyTorch, TensorFlow, Numpy, OpenCV, Pandas, Hugging Face
- Natural Language Processing (NLP):** Langchain, LangGraph, RAG, Meta LLAMA, OpenAI (GPT), Google Gemini

PROJECTS

NavVision: AI-Powered Navigation System for Visually Impaired & Blind People

Tech Stack: YOLOv8n, ZoeD-M12-NK, Gemma, LangChain, Flutter, Python

- Built a real-time assistive navigation system using YOLOv8n (24-class custom fine-tuned model on 20K+ images), achieving up to 83% accuracy on key object classes.
- Designed a multimodal AI pipeline combining YOLO-based object detection, ZoeD-M12-NK monocular depth estimation, and Gemma-based scene understanding for environment awareness.
- Engineered optimized inference pipeline with FP16 precision and 640×480 resolution, achieving ~15 FPS for detection and ~1 FPS for depth estimation on edge devices.
- Developed a mobile-first Flutter + Python architecture enabling real-time navigation assistance without external sensors, supporting offline intelligent perception.

Shariah Law Assistant: AI-Powered Multilingual Islamic Q&A System

Tech Stack: FastAPI, Streamlit, LangChain, ChromaDB, Gemini 1.5 Flash, HuggingFace Transformers

- Built a Shariah-based RAG Q&A system using Quran, Hadith, and classical Islamic texts, enabling context-aware responses.
- Developed a multilingual embedding pipeline using paraphrase-multilingual-mpnet-base-v2, supporting Arabic and cross-language queries with semantic search capabilities.
- Implemented a FastAPI backend with LangChain orchestration and ChromaDB vector store, integrating Gemini 1.5 Flash for responses.
- Designed a lightweight Streamlit frontend for interactive querying and real-time knowledge retrieval.

Finviz Consultants: Corporate Service Platform

Tech Stack: Next.js, TypeScript, Supabase, Framer Motion, Tailwind CSS

- Developed a full-stack platform using Next.js with separate admin and user portals for structured service workflows.
- Integrated Supabase for authentication, database management, and real-time data handling.
- Enhanced UI/UX using Framer Motion animations and designed modular frontend architecture using SSR, visit [here](#).

SK Sports: Full-Stack E-Commerce Sports Equipment Store

Tech Stack: Redux Toolkit, React.js, Node Js, EmailJS, MongoDB, Cloudinary, Render, Tailwind CSS

- Built a full-stack e-commerce platform using MERN with cart, guest checkout, order tracking, and admin dashboard functionality.
- Implemented Redux Toolkit for scalable state management across user and admin workflows.
- Integrated EmailJS for automated order communication and updates.
- Deployed production-ready application on Render with custom domain configuration, visit [here](#).

CERTIFICATIONS

- Advanced Learning Algorithms – DeepLearning.AI (verification: coursera.org/account/accomplishments/certificate/EWR0UCLLCPDK)
- Supervised Machine Learning – DeepLearning.AI (verification: coursera.org/account/accomplishments/certificate/GE6PVMRHZOYS)